Lecture # 14

Thursday
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10-11 AM
3-4 PM

PH102
After Mid Semester Exams

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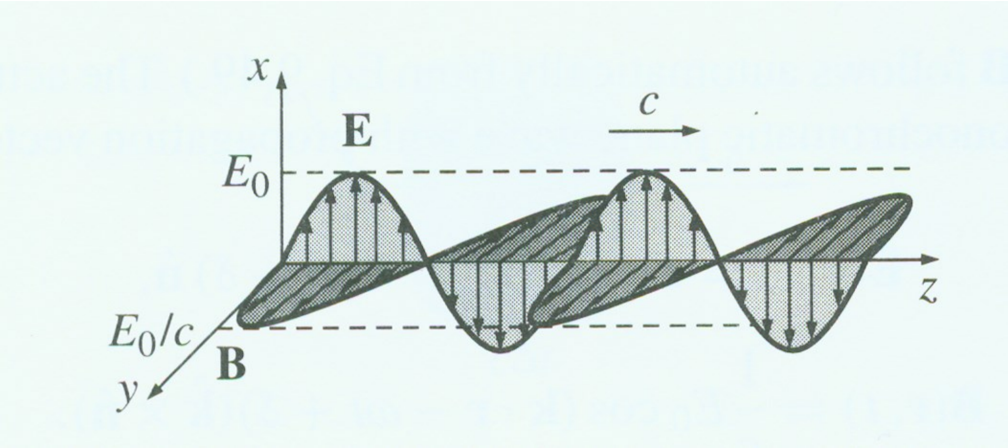
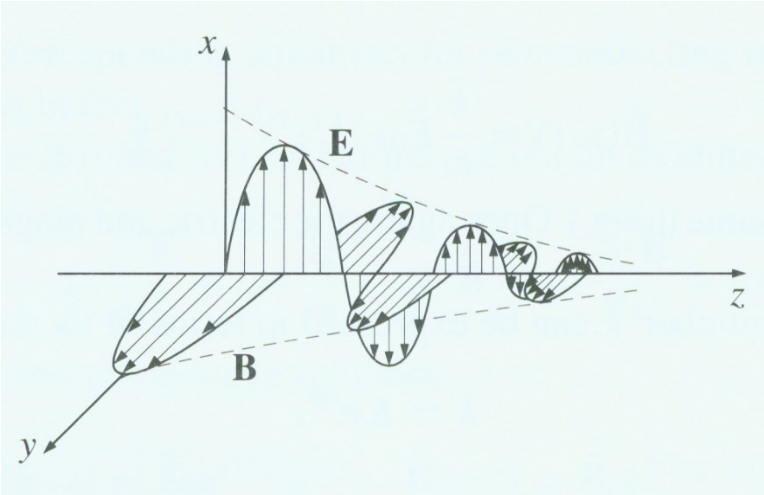
Chapter-9
EM Waves in Conducting Medium

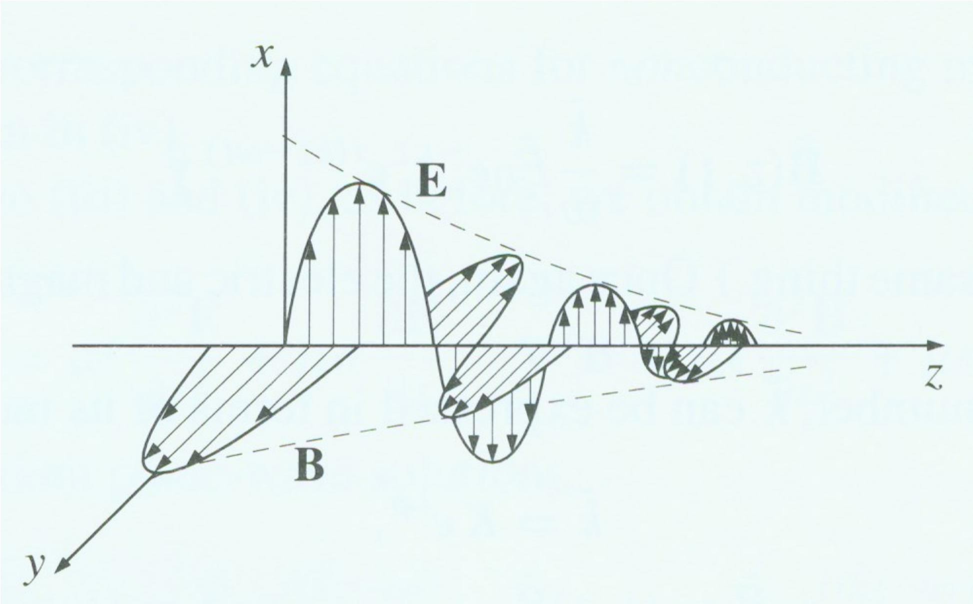
EM Waves in Conducting Medium

Vacuum



Conductor





For a conducting medium

$$\vec{J} = \sigma \vec{E}$$

Maxwell's equations become:

$$\vec{\nabla} \cdot \vec{E} = 0 \longrightarrow (i)$$

$$\vec{\nabla} \cdot \vec{H} = 0 \longrightarrow (ii)$$

$$\vec{\nabla} \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t} \longrightarrow (iii)$$

$$\vec{\nabla} \times \vec{H} = \sigma \vec{E} + \varepsilon \frac{\partial \vec{E}}{\partial t} \longrightarrow (iv)$$

Taking curl of (iii) we obtain

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\mu \frac{\partial}{\partial t} (\vec{\nabla} \times \vec{H})$$

$$\Rightarrow \vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - \nabla^2 \vec{E} = -\mu\sigma \frac{\partial \vec{E}}{\partial t} - \mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\Rightarrow \nabla^2 \vec{E} - \mu\sigma \frac{\partial \vec{E}}{\partial t} - \mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2} = 0$$

EM Waves in Conducting Medium

$$\vec{E} = \hat{x} E_0 e^{i(kz - \omega t)}$$

Then, using $\vec{\nabla} \rightarrow i\vec{K}$

$\frac{\partial}{\partial t} \rightarrow -i\omega$

we can write

$$-k^2 + i\mu\sigma\omega + \omega^2\mu\varepsilon = 0$$

clearly k is a complex number.

$$K = \alpha + i\beta$$

Phase constant

Attenuation constant

$$-\left(\alpha^2 - \beta^2 + 2i\alpha\beta\right) + i\omega\mu\sigma + \omega^2\varepsilon\mu = 0$$

Equating real and imaginary parts, we obtain

$$\alpha^2 - \beta^2 = \omega^2\varepsilon\mu$$

$$2\alpha\beta = \omega\mu\sigma$$

$$\Rightarrow \beta = \frac{\omega\mu\sigma}{2\alpha}$$

Thus, $$\alpha^2 - \frac{\omega^2 \mu^2 \sigma^2}{4\alpha^2} = \omega^2 \varepsilon \mu$$

$$\Rightarrow \quad \alpha^4 - \omega^2 \mu \varepsilon \alpha^2 - \frac{1}{4}\omega^2 \mu^2 \sigma^2 = 0$$

$$\Rightarrow \quad \alpha^2 = \frac{\omega^2 \mu \varepsilon \pm \sqrt{\omega^4 \mu^2 \varepsilon^2 + \omega^2 \mu^2 \sigma^2}}{2}$$

$$\Rightarrow \quad \alpha^2 = \frac{\omega^2 \mu \varepsilon \pm \omega^2 \mu \varepsilon \sqrt{1 + \sigma^2/\omega^2 \varepsilon^2}}{2}$$

$$\Rightarrow \quad \alpha = \sqrt{\omega^2 \mu \varepsilon} \left[\frac{1}{2} \pm \frac{1}{2}\left(1 + \frac{\sigma^2}{\omega^2 \varepsilon^2}\right)^{1/2} \right]^{1/2}$$

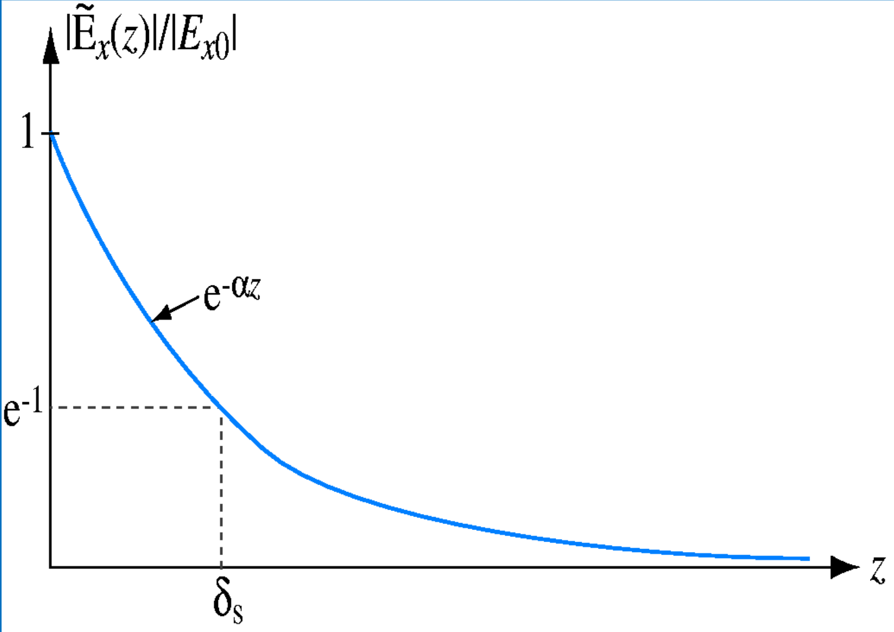
as α is real we choose +ve sign!

Thus,

$$\alpha = \omega\sqrt{\varepsilon\mu}\left[\frac{1}{2} + \left(1 + \frac{\sigma^2}{\omega^2\varepsilon^2}\right)^{1/2}\right]^{1/2}$$

$$\beta = \frac{\omega\mu\sigma}{2\alpha}$$

When κ is complex we have



Skin Depth, δ_s

shows how well an electromagnetic wave can penetrate into a conducting medium

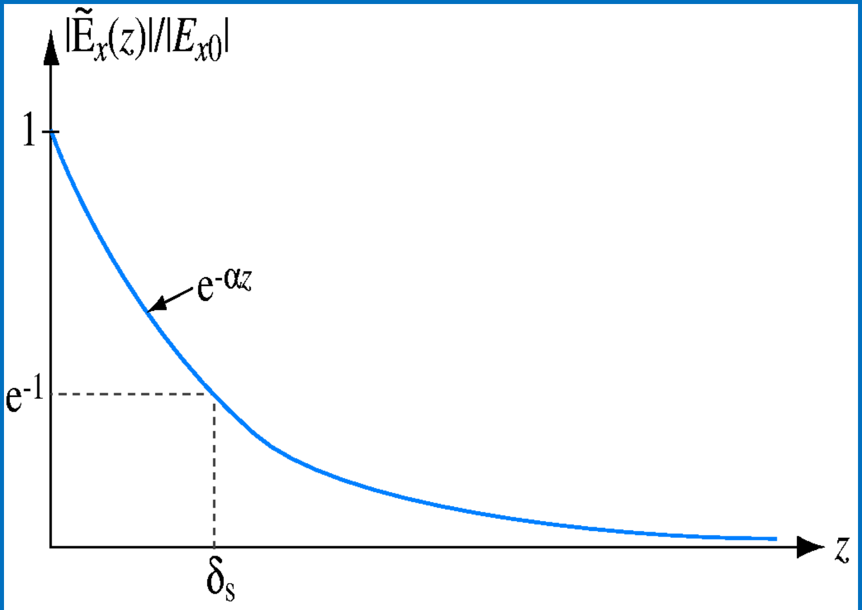
Skin Depth δ_s : [m]

Perfect dielectric:

$\sigma=0$ $\alpha=0$ $\delta_s=\infty$

Perfect Conductor:

$\sigma=\infty$ $\alpha=\infty$ $\delta_s=0$



$$\vec{E} = \vec{E_0}\, e^{-\beta z}\, e^{i(\alpha z - \omega t)} \longrightarrow (1)$$

$\Rightarrow$ The EM wave gets attenuated while propagating through a conducting medium.

The attenuation is due to Joule loss. For a good conductor $\frac{\sigma}{\omega\varepsilon} \gg 1$

and we obtain

$$\alpha = \omega\sqrt{\varepsilon\mu}\left[\frac{1}{2} + \frac{1}{2}\left(\cancel{1+}\;\frac{\sigma}{\omega\varepsilon}\right]^{1/2}$$

$$= \omega\sqrt{\frac{\varepsilon\mu}{2}}\sqrt{\frac{\sigma}{\omega\varepsilon}}$$

$$= \sqrt{\frac{\omega\mu\sigma}{2}}$$

$$\beta = \frac{\omega\mu\sigma}{2\alpha} = \frac{\sigma\omega\mu\sqrt{2}}{2\sqrt{\omega\mu\sigma}}$$

$$= \sqrt{\frac{\omega\mu\sigma}{2}}$$

$$\alpha \simeq \beta = \sqrt{\frac{\omega\mu\sigma}{2}}$$

If $\frac{\sigma}{\omega\varepsilon} << 1$ the medium is classified as a cond dielectric.

For

$$0.01 \leq \frac{\sigma}{\omega \varepsilon} \leq 100$$

the medium is said to be a quasi-conductor. Note from (1) that the field decay by a factor e in traversing a distance $\delta = \frac{1}{\beta}$ which is known as the penetration depth.

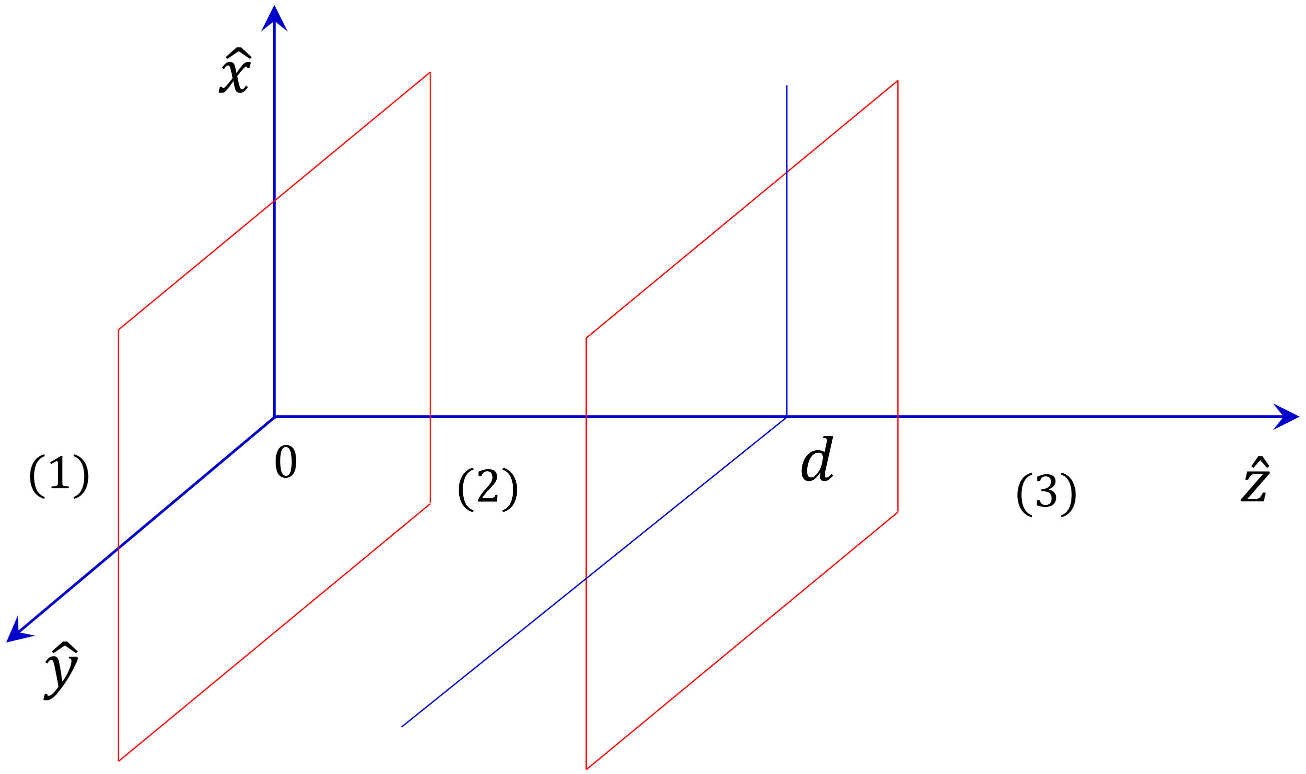
Example:

Light of (angular) frequency ω passes from medium 1, through a slab (thickness d) of medium 2, and into medium 3. Show that the transmission coefficient for normal incidence is given by,

$$T^{-1} = \frac{1}{4n_1n_3}\left[(n_1 + n_3)^2 + \frac{({n_1}^2 - {n_2}^2)({n_3}^2 - {n_2}^2)}{{n_2}^2} sin^2\left(\frac{n_2\omega d}{c}\right)\right]$$

{Hint: To the left, there is an incident wave and a reflected wave; to the right, there is a transmitted wave; inside the slab there is a wave going to the right and a wave going to the left. Express each of these in terms of its complex amplitude, and relate the amplitudes by imposing suitable boundary conditions at the two interfaces. All three media are linear and homogeneous; assume $\mu_1 = \mu_2 = \mu_3 = \mu_0$

Solution :-



$z < 0$ Incident EM waves are

$$\begin{cases} \widetilde{\boldsymbol{E}_{\mathrm{I}}}(z,t) = \widetilde{E}_{\mathrm{I}} e^{i(k_1 z - \omega t)} \hat{x} \\ \widetilde{\boldsymbol{E}_R}(z,t) = \widetilde{E_R} e^{i(-k_1 z - \omega t)} \hat{x} \end{cases}$$

$$\begin{cases} \widetilde{\boldsymbol{B}_{\mathrm{I}}}(z,t) = \dfrac{1}{v_1} \widetilde{E}_{\mathrm{I}} e^{i(k_1 z - \omega t)} \hat{y} \\ \widetilde{\boldsymbol{B}_R}(z,t) = -\dfrac{1}{v_1} \widetilde{E_R} e^{i(-k_1 z - \omega t)} \hat{y} \end{cases}$$

$0 < z < d$ *EM* wave is:

$$\begin{cases} \widetilde{\boldsymbol{E}_r}(z,t) = \widetilde{E_r} e^{i(k_2 z - \omega t)} \hat{x} \\ \\ \widetilde{\boldsymbol{E}_l}(z,t) = \widetilde{E}_l e^{i(-k_2 z - \omega t)} \hat{x} \end{cases}$$

$$\begin{cases} \widetilde{\boldsymbol{B}_r}(z,t) = \frac{1}{v_2} \widetilde{E_r} e^{i(k_2 z - \omega t)} \hat{y} \\ \\ \widetilde{\boldsymbol{B}_l}(z,t) = -\frac{1}{v_2} \widetilde{E}_l e^{i(-k_2 z - \omega t)} \hat{y} \end{cases}$$

$z > d$ Transmitted *EM* wave is

$$\widetilde{\boldsymbol{E}_T}(z,t) = \widetilde{E_T} e^{i(k_3 z - \omega t)} \hat{x}$$

$$\widetilde{\boldsymbol{B}_T}(z,t) = \frac{1}{v_3} \widetilde{E_T} e^{i(k_3 z - \omega t)} \hat{y}$$

Now applying the boundary condition

$$\boxed{\mathrm{E}_1^{\parallel} = \mathrm{E}_2^{\parallel}} \qquad \boxed{\mathrm{B}_1^{\parallel} = \mathrm{B}_2^{\parallel}}$$

At each boundary

Given $\mu_1 = \mu_2 = \mu_3 = \mu_0$

At z = 0

$$\widetilde{E}_I + \widetilde{E}_R = \widetilde{E}_r + \widetilde{E}_l \qquad (i)$$

$$\frac{\widetilde{E}_I}{v_1} - \frac{\widetilde{E}_R}{v_1} = \frac{\widetilde{E}_r}{v_2} - \frac{\widetilde{E}_l}{v_2} \Rightarrow \widetilde{E}_I - \widetilde{E}_R = \frac{v_1}{v_2}(\widetilde{E}_r - \widetilde{E}_l)$$

$$\widetilde{E}_I - \widetilde{E}_R = \beta\,(\widetilde{E}_r - \widetilde{E}_l) \qquad (ii)$$

Where $\beta = \frac{v_1}{v_2}$

At z = d

$$\widetilde{E}_r e^{ik_2 d} + \widetilde{E}_l e^{-ik_2 d} = \widetilde{E}_T e^{ik_3 d} \qquad (iii)$$

$$\frac{1}{v_2}\widetilde{E}_r e^{ik_2 d} - \frac{1}{v_2}\widetilde{E}_l e^{-ik_2 d} = \frac{1}{v_3}\widetilde{E}_T e^{ik_3 d}$$

$$\widetilde{E_r}e^{ik_2d} - \widetilde{E}_l e^{-ik_2d} = \frac{v_2}{v_3}\widetilde{E_T}e^{ik_3d}$$

$$\boxed{\widetilde{E_r}e^{ik_2d} - \widetilde{E}_l e^{-ik_2d} = \alpha\widetilde{E_T}e^{ik_3d}} \quad (iv)$$

Where $\alpha = \frac{v_2}{v_3}$

Adding equation (i) & (ii) we get

$$\boxed{2\widetilde{E}_{\mathrm{I}} = \widetilde{E_r}(1+\beta) + \widetilde{E}_l(1-\beta)} \quad (v)$$

Adding equation (iii) & (iv) we get

$$\boxed{2\widetilde{E_r}e^{ik_2d} = (1+\alpha)\widetilde{E_T}e^{ik_3d}}$$

Subtract equation (iii) & (iv)

$$\boxed{2\widetilde{E}_l e^{-ik_2 d} = (1-\alpha)\widetilde{E_T} e^{ik_3 d}}$$

Substitute value of $\widetilde{E_r}$ & $\widetilde{E}_l$ into equation (v)

$$2\widetilde{E}_\mathrm{I} = \frac{1}{2}(1+\beta)(1+\alpha)e^{-ik_2 d}\widetilde{E_T} e^{ik_3 d} + \frac{1}{2}(1-\beta)(1-\alpha)e^{ik_2 d}\widetilde{E_T} e^{ik_3 d}$$

$$4\widetilde{E}_\mathrm{I} = \left[(1+\beta)(1+\alpha)e^{-ik_2 d} + (1-\beta)(1-\alpha)e^{ik_2 d}\right]\widetilde{E_T} e^{ik_3 d}$$

$$4\widetilde{E}_\mathrm{I} = \left[(1+\alpha\beta)(e^{-ik_2 d} + e^{ik_2 d}) + (\alpha+\beta)(e^{-ik_2 d} - e^{ik_2 d})\right]\widetilde{E_T} e^{ik_3 d}$$

$$\boxed{4\widetilde{E}_\mathrm{I} = 2[(1+\alpha\beta)cosk_2 d - i(\alpha+\beta)sink_2 d]\widetilde{E_T} e^{ik_3 d}}$$

Transmission coefficient is

$$T = \frac{v_3 \epsilon_3 {E_T}^2}{v_1 \epsilon_1 {E_\mathrm{I}}^2}$$

$$T = \frac{v_3}{v_1}\left(\frac{\mu_0 \epsilon_3}{\mu_0 \epsilon_1}\right)\frac{|E_T|^2}{|E_\mathrm{I}|^2}$$

$$T = \frac{v_1}{v_3}\frac{{E_T}^2}{{E_\mathrm{I}}^2} \Rightarrow \boxed{T = \alpha\beta \frac{|E_T|^2}{|E_\mathrm{I}|^2}}$$

$$T^{-1} = \frac{1}{\alpha\beta}\frac{|E_\mathrm{I}|^2}{|E_T|^2}$$

$$= \frac{1}{\alpha\beta}\left|\frac{1}{2}[(1+\alpha\beta)cosk_2 d - i(\alpha+\beta)sink_2 d]e^{ik_3 d}\right|^2$$

$$T^{-1} = \frac{1}{4\alpha\beta}[(1+\alpha\beta)^2 cos^2 k_2 d + (\alpha+\beta)^2 sin^2 k_2 d]$$

$$T^{-1} = \frac{1}{4\alpha\beta}[(1+\alpha\beta)^2 (1 - sin^2 k_2 d) + (\alpha+\beta)^2 sin^2 k_2 d]$$

$$T^{-1} = \frac{1}{4\alpha\beta}[(1+\alpha\beta)^2 + (1-\alpha^2)(1-\beta^2\,) sin^2 k_2 d]$$

Where $\beta = \frac{v_1}{v_2} = \frac{n_2}{n_1}$

$\alpha = \frac{v_2}{v_3} = \frac{n_3}{n_2}$

But $n_1 = \frac{c}{v_1}$, $n_2 = \frac{c}{v_2}$, $n_3 = \frac{c}{v_3}$

And $k_2 = \frac{n_2 \omega}{c}$

Therefore

$$T^{-1} = \frac{1}{4n_1 n_3}\left[(n_1+n_3)^2 + \frac{({n_1}^2 - {n_2}^2)({n_3}^2 - {n_2}^2)}{{n_2}^2} sin^2\left(\frac{n_2 \omega d}{c}\right)\right]$$

Example:

The magnetic field component of a plane wave in a lossless dielectric is
$\vec{H} = 30sin(2\pi \times 10^8 t - 5x)\hat{z}$ mA/m

(a) If $\mu_r = 1$ find ε_r .
(b) Calculate the wavelength and wave velocity.
(c) Determine the wave impedance.
(d) Determine the polarization of the wave.
(e) Find the corresponding electric field component.
(f) Find the displacement current density.

Solution :- (a) Given $\vec{H} = 30sin(2\pi \times 10^8 t - 5x)\hat{z}$

$$\omega = 2\pi \times 10^8 \ \& \ \mathrm{k} = 5$$

We know that

$$\boxed{\omega = kv = \frac{k}{\sqrt{\mu\epsilon}}}$$

And

$$\sqrt{\mu\epsilon} = \sqrt{\mu_r\epsilon_r}\frac{1}{c}$$

$$\boxed{\sqrt{\mu\epsilon} = \sqrt{\epsilon_r}\frac{1}{c}}$$

[As $\mu_r = 1$]

So,

$$\boxed{\omega = \frac{k}{\sqrt{\mu\epsilon}} = \frac{kc}{\sqrt{\epsilon_r}}} \Longrightarrow \boxed{\sqrt{\epsilon_r} = \frac{kc}{\omega} = \frac{(5)(3\times 10^8)}{2\pi\times 10^8} = \frac{15}{2\pi}}$$

$$\Downarrow$$

$$\boxed{\epsilon_r = 5.705}$$

(b) Wavelength $\lambda = \frac{2\pi}{k}$

$$\boxed{\lambda = \frac{2\pi}{5} = 1.256\, m}$$

Wave velocity

$$\boxed{v = \frac{c}{\sqrt{\mu_r\epsilon_r}} = 1.256\times 10^8\, m/sec}$$

(c) Wave impedance is ,

$$Z = \sqrt{\frac{\mu}{\varepsilon}}$$

$$= Z\sqrt{\frac{\mu_0\mu_r}{\epsilon_0\epsilon_r}} = \frac{\mu_0 c}{\sqrt{\epsilon_r}}$$

$$Z = \frac{\mu_0 c}{\sqrt{\epsilon_r}} = \frac{(4\pi \times 10^{-7})(3 \times 10^8)}{15/2\pi} \implies \boxed{Z = 157.75\Omega}$$

(d) Direction of magnetic field(**H**) is along $\hat{z}$ and wave propagation vector along $\hat{x}$

$$\hat{n} = \hat{z} \times \hat{x} = \hat{y}$$

$\implies$ direction of electric field (**E**) along $\hat{y}$

Determine the polarization of the wave is along $\hat{y}$

(e)Now we determine Electric field component

Impedance $$Z = \frac{E}{H} \Longrightarrow \boxed{E = ZH}$$

$$\Downarrow$$

$$E = (157.75)(30 \times 10^{-3} sin(2\pi \times 10^8 t - 5x))$$

$$\boxed{\vec{E} = 4.732\, sin(2\pi \times 10^8 t - 5x)\hat{y} \text{ V/m}}$$

(f) Displacement current density.

$$J_d = \frac{\partial \vec{D}}{\partial t} = \nabla \times \vec{H}$$

$$J_d = \nabla \times \vec{H} = \nabla \times \{30 \times 10^{-3} sin(2\pi \times 10^8 t - 5x)\hat{z}\}$$

$$\boxed{J_d = 0.15\, sin(2\pi \times 10^8 t - 5x)\hat{y}}$$

Example:

An EM wave travels in free space with the electric field component

$\vec{E} = 100e^{j(0.866y+0.5z)}\hat{x}$ V/m

Determine

(a) ω and λ

(b) The magnetic field component .

(c) The time average power in the wave .

Solution :-

(a) Given $\vec{E} = 100e^{j(0.866y+0.5z)}\hat{x}$ V/m

Comparing the given E with

$$\vec{E} = E_0 e^{j\boldsymbol{k}.\boldsymbol{r}}\hat{x} \qquad \vec{E} = E_0 e^{j(k_x x+k_y y+k_z z)}\hat{x}$$

We get $k_x = 0 \qquad k_x = 0.866 \qquad k_x = 0.5$

Thus

$$k = \sqrt{{k_x}^2 + {k_y}^2 + {k_z}^2}$$

$$k = 0.999 \approx 1$$

In free space

$$\boxed{\omega = kc = 3 \times 10^8\, rad/sec}$$

Wavelength $\lambda = \frac{2\pi}{k} = 2\pi = 6.28$

(b) The magnetic field

$$\vec{H} = \frac{\vec{k} \times \vec{E}}{\mu_0 \omega}$$

$$\vec{H} = \frac{(0.866\hat{y} + 0.5\hat{z}) \times (100e^{j(0.866y+0.5z)}\hat{x})}{(4\pi \times 10^{-7})(3 \times 10^8)}$$

$$\vec{H} = \frac{(0.866\hat{y} + 0.5\hat{z}) \times (100e^{j(0.866y+0.5z)}\hat{x})}{(4\pi \times 10^{-7})(3 \times 10^{8})}$$

$$\boxed{\vec{H} = (1.33\hat{y} - 2.3\hat{z})e^{j(0.866y+0.5z)}}$$

(c) Poynting vector $\vec{S} = \vec{E} \times \vec{H}$

$$\vec{S} = \vec{E} \times \left(\frac{\vec{k} \times \vec{E}}{\mu_0 \omega}\right)$$

Use vector triple product

$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A}.\vec{C}) - \vec{C}(\vec{A}.\vec{B})$

$$\vec{S} = \frac{\vec{k}E^2}{\mu_0 \omega}$$

The time average power in the wave

$$\langle P \rangle = 1/2(\vec{E} \times \vec{H})$$

$$\langle P \rangle = \frac{1}{2}\frac{\vec{k}E^2}{\mu_0 \omega}$$

$$\langle P \rangle = \frac{1}{2}\frac{(0.866\hat{y} + 0.5\hat{z})(100)^2}{(4\pi \times 10^{-7})(3 \times 10^8)}$$

$$\langle P \rangle = \frac{1}{2}\frac{(0.866\hat{y} + 0.5\hat{z})(100)^2}{(4\pi \times 10^{-7})(3 \times 10^8)}$$

$$\boxed{\langle P \rangle = 11.49\hat{y} + 6.63\hat{z} \ \mathrm{W/m^2}}$$

Example:

In free space $(z < 0)$, a plane wave with

$$\vec{H} = 10\cos(10^8 t - kz)\,\hat{x}mA/m$$

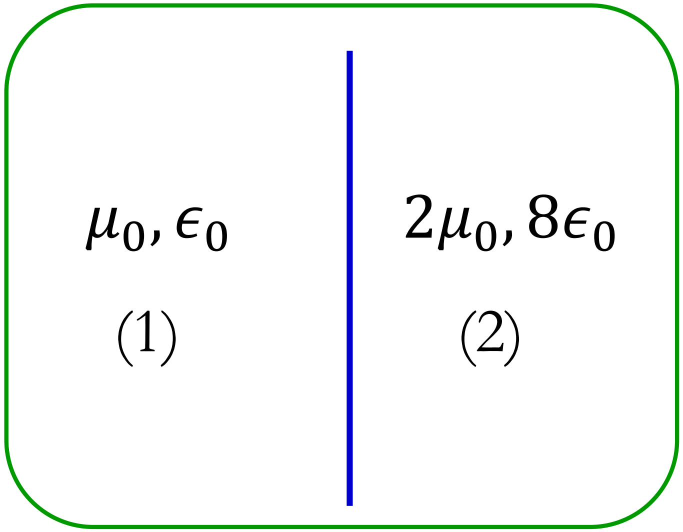
is incident normally on a lossless medium $(\epsilon = 2\epsilon_0, \mu = 8\mu_0)$ in region $z > 0$. Determine the reflected wave H_r , E_r and the transmitted wave H_t, E_t

Solution :- Given $\vec{H} = 10\cos(10^8 t - kz)\,\hat{x}$

We know that $\boxed{\omega = kv}$

$$\Downarrow$$

In free space $\boxed{k_1 = \frac{\omega}{c}}$



$$\boxed{k_1 = \frac{10^8}{3 \times 10^8} = \frac{1}{3}}$$

In medium (1)

Impedance of free space (medium (1))

$$\boxed{Z_1 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 376.63}$$

For the lossless dielectric medium (2),

$$\boxed{\omega = k_2 v}$$

$$\boxed{k_2 = \frac{\omega}{v} = \omega\sqrt{\mu\epsilon} = \frac{\omega}{c}4 = 4k_1 = \frac{4}{3}}$$

Impedance of medium (2)

$$\boxed{Z_2 = \sqrt{\frac{\mu}{\epsilon}} = 2Z_1 = 753.26}$$

Given $\overrightarrow{H_i} = 10\cos(10^8 t - k_1 z)\,\hat{x}$

We expect that $\overrightarrow{E_i} = E_{i0}\cos(10^8 t - k_1 z)\,\hat{n}$

Where , $\hat{n} = \hat{x} \times \hat{z} = -\hat{y}$

And $E_{i0} = Z_1 H_{i0} = (10) Z_1$

Hence $\overrightarrow{E_i} = 10 Z_1 \cos\left(10^8 t - \frac{1}{3} z\right)(-\hat{y})$

$$\boxed{E_{r0} = \frac{1-\beta}{1+\beta} E_{i0}}$$

Griffith Eq^n : 9.82

Where , $\beta = \frac{\mu_1 v_1}{\mu_2 v_2}$

$$= \frac{\mu_0 c}{(8\mu_0)\left(\frac{c}{4}\right)} = \frac{1}{2}$$

$$\Downarrow$$

$$E_{r0} = \frac{1}{3} E_{i0}$$

So,

$$\boxed{\overrightarrow{E_r} = \frac{10}{3} Z_1 \cos\left(10^8 t + \frac{1}{3} z\right)(-\hat{y}) \text{ mV/m}}$$

$$\boxed{\overrightarrow{H_r} = \frac{\overrightarrow{E_r}}{Z_1} = -\frac{10}{3} \cos\left(10^8 t + \frac{1}{3} z\right) \hat{x}\ mA/m}$$

We know

$$\boxed{E_{t0} = \frac{2}{1+\beta} E_{i0}}$$

Griffith Eq^{n}: 9.82

$$\beta = \frac{\mu_0 c}{(8\mu_{0)}(\frac{c}{4})} = \frac{1}{2}$$

$$\boxed{E_{t0} = \frac{4}{3} E_{i0}}$$

And $E_{i0} = (10)Z_1$

$$\overrightarrow{E_t} = E_{t0} \cos(10^8 t - k_2 z)\,\hat{n}$$

$$\boxed{\overrightarrow{E_t} = -\frac{40}{3} Z_1 \cos\left(10^8 t - \frac{4}{3} z\right) \hat{y} \text{ mV/m}}$$

And

$$\boxed{\overrightarrow{H_t} = \frac{\overrightarrow{E_t}}{Z_2} = \frac{20}{3} \cos\left(10^8 t - \frac{4}{3} z\right) \hat{x} \text{ mA/m}}$$

$$\boxed{Z_2 = 2Z_1}$$